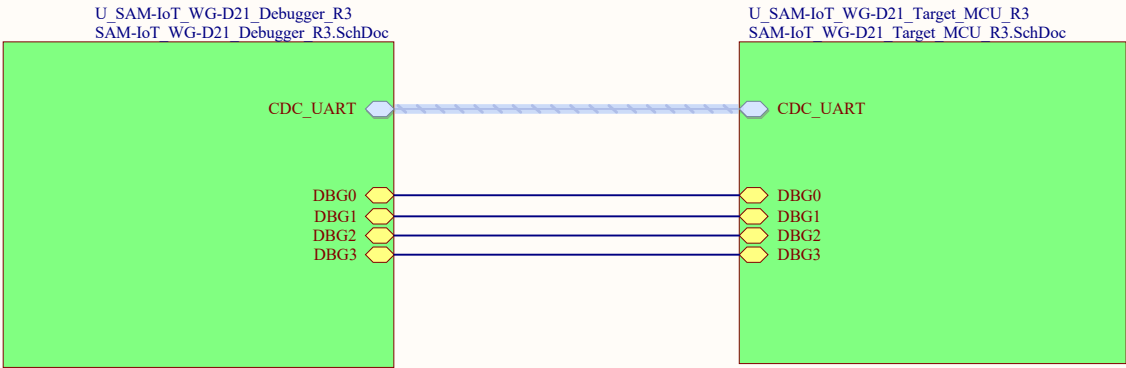


SAM-IoT WG-D21



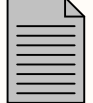
A11-0463

A12-1410

FW1

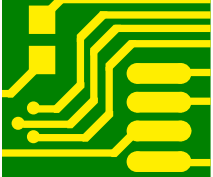


PCBADOC1



A12-1409

A11-0406



A08-3035
SAM-IoT WG-D21 PCB documentation

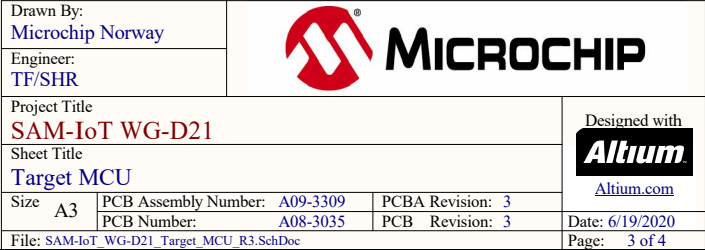
A09-0612



Label PCBA Small 6x6mm

U_SAM-IoT_WG-D21_Revision_History_R3
SAM-IoT_WG-D21_Revision_History_R3.SchDoc

Drawn By: Microchip Norway		 MICROCHIP		
Engineer: TF/SHR				
Project Title SAM-IoT WG-D21				Designed with  Altium.com
Sheet Title Top Level				
Size	A3	PCB Assembly Number: A09-3309	PCBA Revision: 3	
		PCB Number: A08-3035	PCB Revision: 3	
File: SAM-IoT_WG-D21_TopLevel_R3.SchDoc				Date: 6/19/2020
				Page: 1 of 4

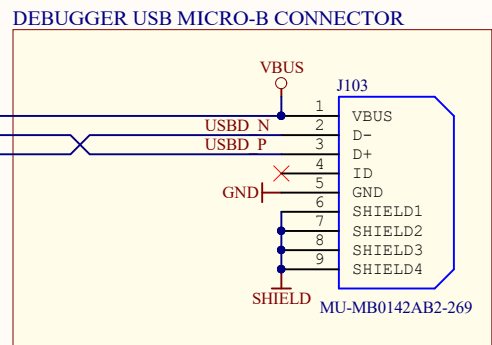


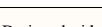
VCC_MUX is powered from two different sources:

1. USB cable 5.0V [4.4V to 5.25V]
2. Li-Ion/Li-Po battery 3.7V [3.0V to 4.20V]

MIC33050:

- Vin: 2.5V to 5.5V
- Vout: Fixed 3.3V
- I_{max}: 600mA



Drawn By: Microchip Norway		 MICROCHIP
Engineer: TF/SHR		
Project Title SAM-IoT WG-D21		Designed with  Altium.com
Sheet Title Debugger		
Size A3	PCB Assembly Number: A09-3309	PCBA Revision: 3
	PCB Number: A08-3035	PCB Revision: 3
File: SAM-IoT WG-D21 Debugger R3.SchDoc		Date: 6/19/2020 Page: 3 of 4

Revision History

PCB Assembly Rev 1:

▲

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

▲

Design Changes:

- Added separate power net for the WINC1500

- Changed pull resistors connected to the WINC1500 from 100K to 1MΩ

- Set I2C pull resistors to not mounted as they are available on click modules

- Swapped PA10 and PA25 to get high drive capabilities on the mikroBUS PWM signal

PCB:

PCB revision 2

PCB Assembly Rev 3:

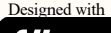
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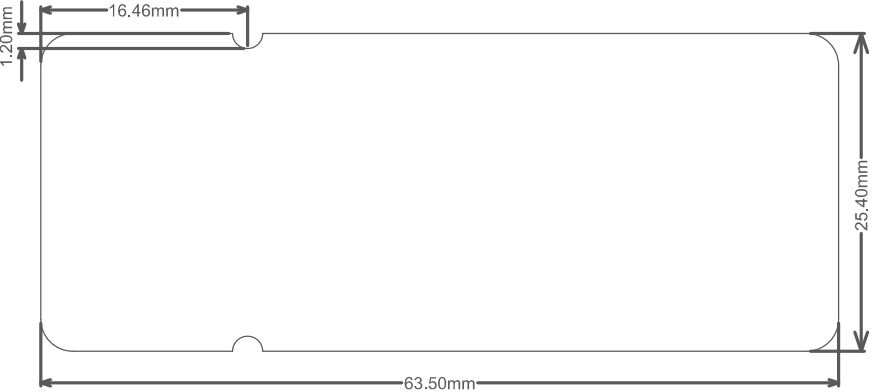
Design Changes:

One signal line was not making good connection. Needed to also be updated in Database.

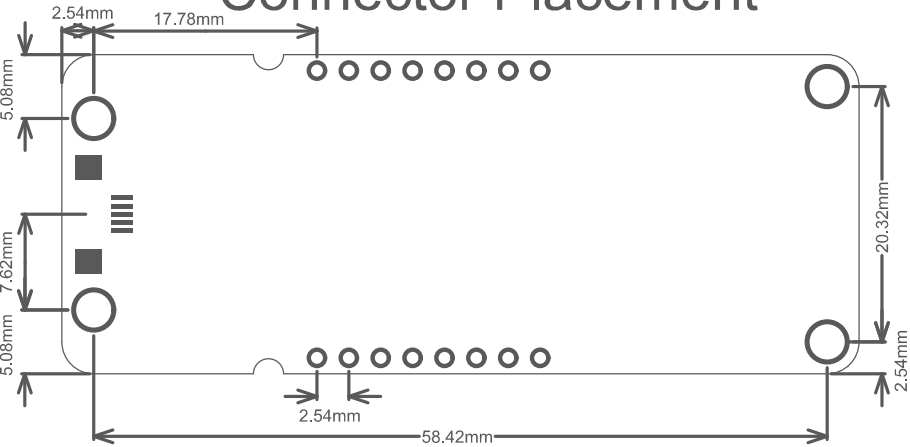
PCB:

PCB revision 2

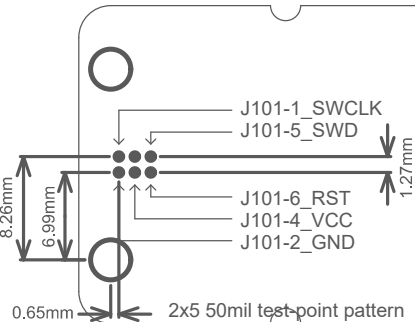
Drawn By: Microchip Norway		 MICROCHIP	
Engineer: TF/SHR			
Project Title SAM-IoT WG-D21			Designed with  Altium.com
Sheet Title Revision History			
Size A3	PCB Assembly Number: A09-3309	PCBA Revision: 3	
	PCB Number: A08-3035	PCB Revision: 3	Date: 6/19/2020
File: SAM-IoT_WG-D21_Revision_History_R3.SchDoc			Page: 4 of 4

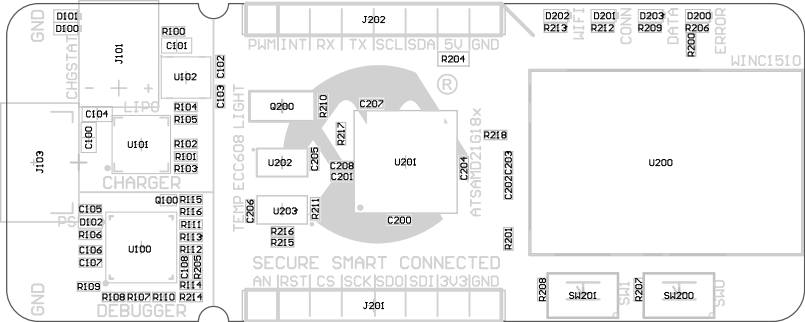


Connector Placement



Test Point Placement





GND

BOOT

J102
TP100

GND

L1P0 -
BATT +



SAMD21

PSU
VCC

DEBUG
401r

TARGET
001r

WINC
501r



PA10 / PWM

AN / PA02

PA24 / INT

RST / PA03

PA09 / RX

CS / PA06

PA08 / TX

SCK / PA05

PA17 / SCL

MISO / PA07

PA16 / SDA

MOSI / PA04

+5V

+3.3V

GND

GND



BUS

GP1

GP3

GP4

GP5

GP6

SCL

SDA

Microchip © 2019

A08-3035 Rev3



TP208

TP207

TP209

TP200 TP201 TP202 TP206 TP205 TP204 TP203

